



Optical Properties 1

Presented by Dr Adijat 'Wumi Inyang

Aims and Learning Outcomes

Describe

Describe the interaction of materials with light at atomic level

Outline

Outline the mechanism of light absorption

Explain

Explain the mechanism of light absorption by materials

Establish

Establish differences between the behavior of different materials in the presence of light

Optical properties

- **Interaction of light with solids**

Light can be:

- Reflected
- Absorbed
- Luminescent
- Transmitted
- Refracted
- Scattered



Optical properties

Optical classification of materials:

- Materials can be classified based on the way they interact with light

Transparent

Translucent

Opaque

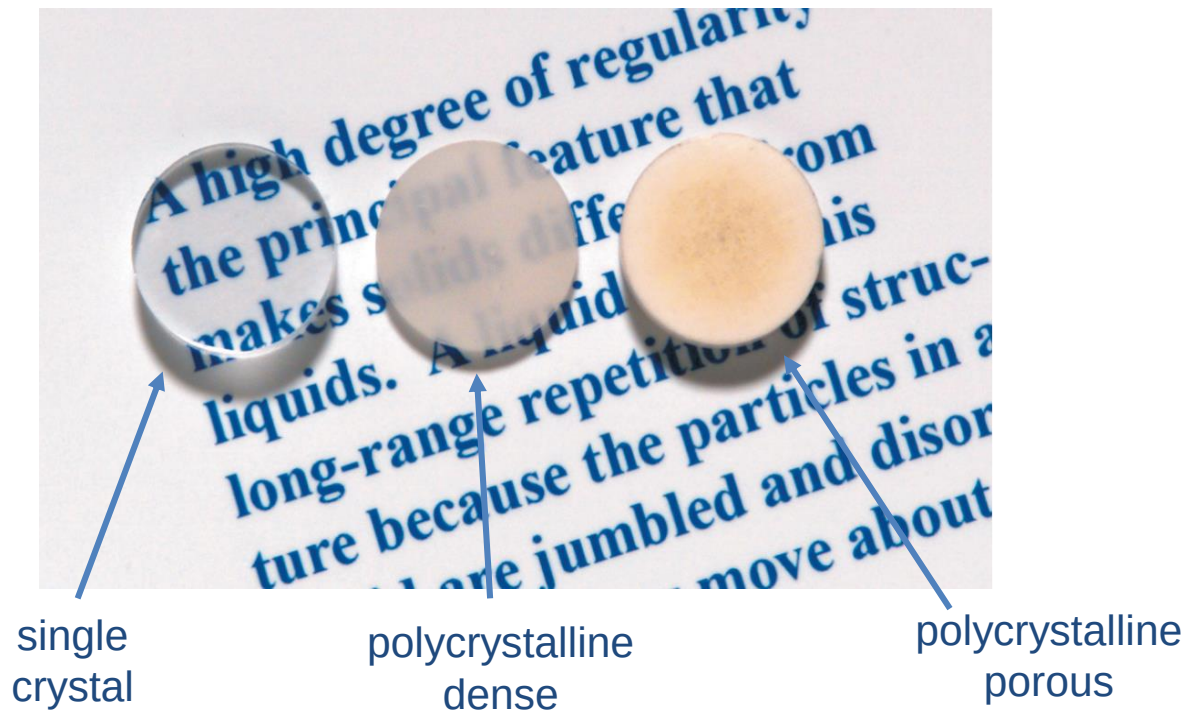
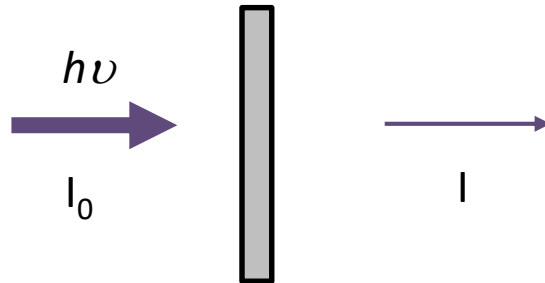


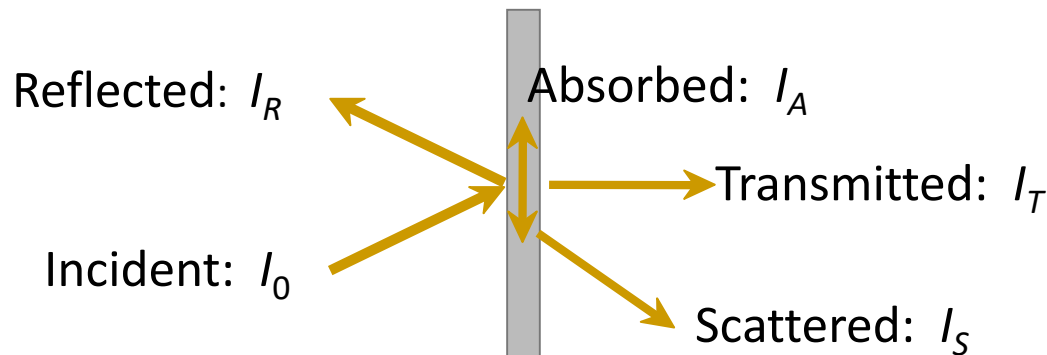
Fig. 21.10, Callister & Rethwisch 9e.

Optical properties

- Light interactions with solids



$$I_0 = I_T + I_A + I_R + I_S$$



Incident light can be:

- T** transmitted
- A** absorbed
- R** reflected
- S** scattered

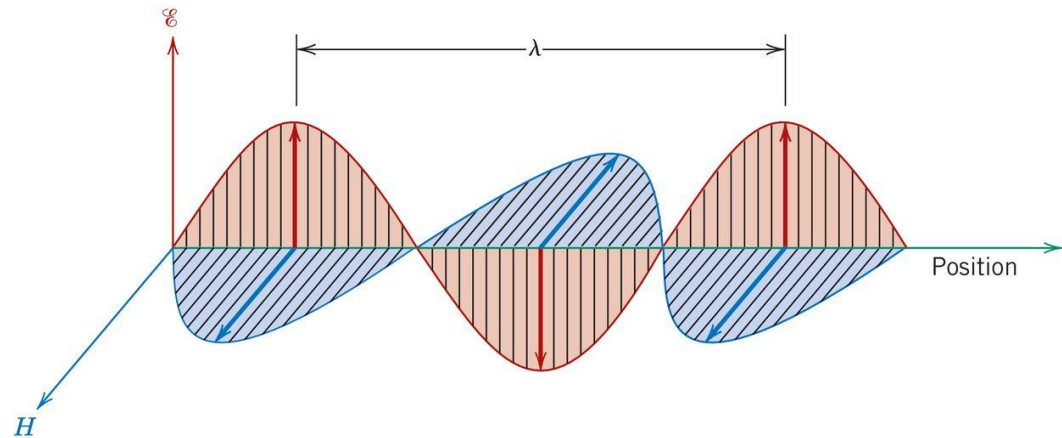
Electro-magnetic radiation

• Interaction of Materials and Radiation

Electro-magnetic radiation

An oscillating electric field generates an magnetic field at 90° to it.

This travels through space at the velocity of light



Wave equation:

$$c = \lambda \nu$$

Light has both particulate and wavelike characteristics

where:

λ = wavelength (m)

ν = frequency (Hz)

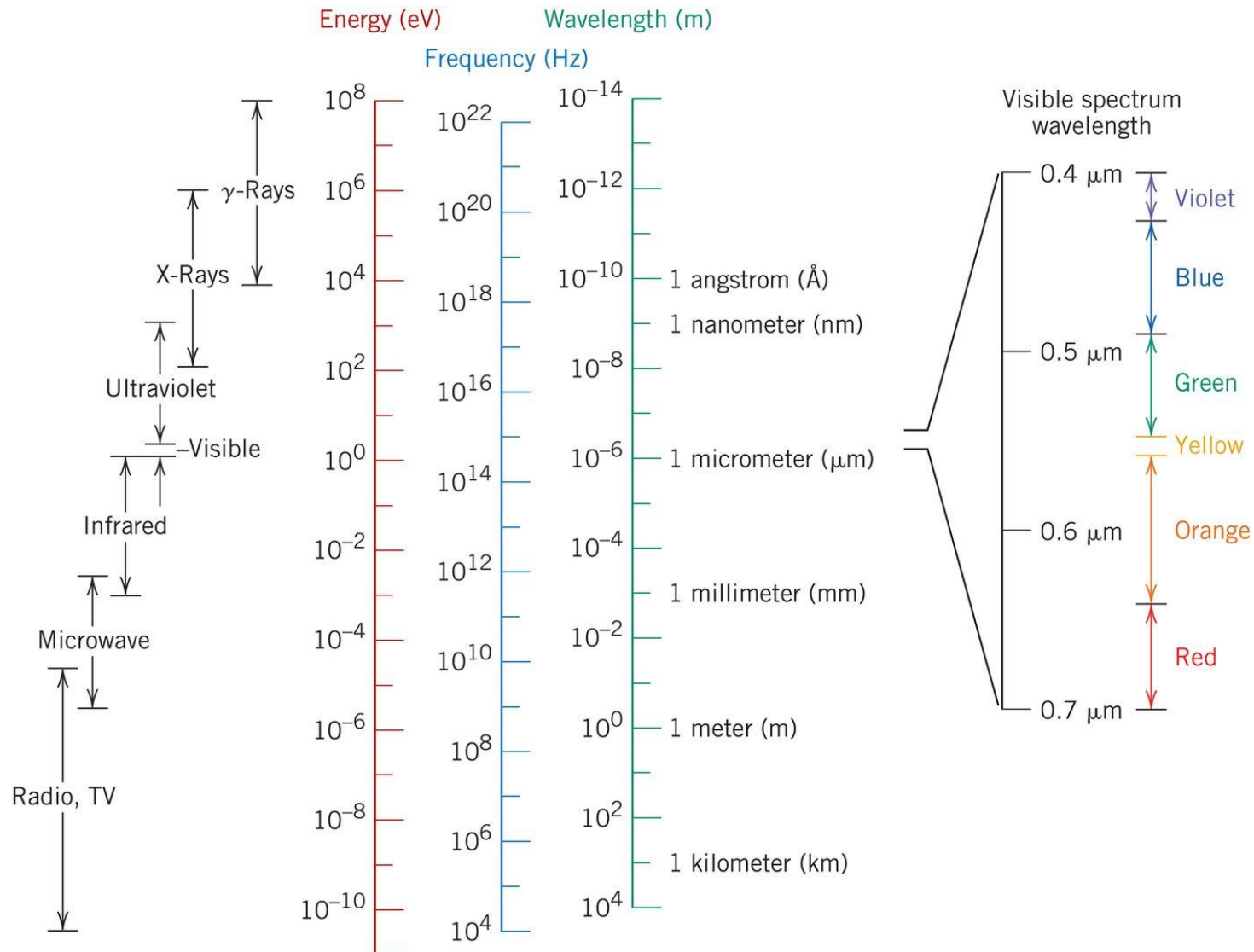
c = velocity of light
2.99x10⁸ m/s

h = Planck's constant
6.626x10⁻³⁴ m²kg/s

- **Photon** - a quantum of light (energy)

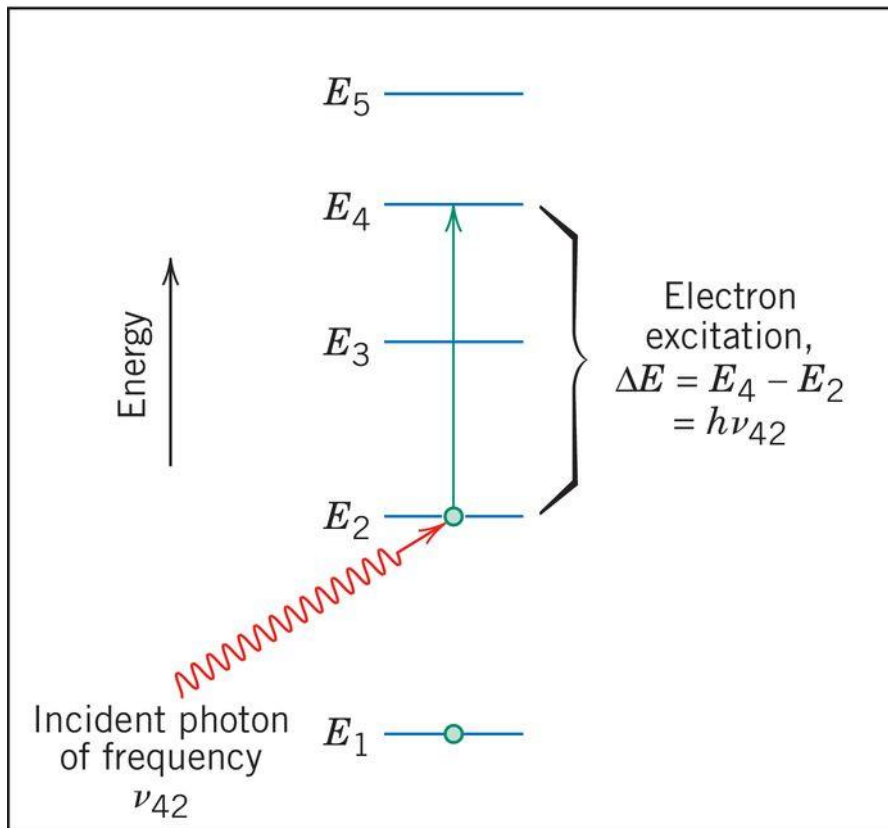
$$E = h\nu = \frac{hc}{\lambda} \quad \text{unit J or eV}$$

Electro-magnetic spectrum



Electron transitions

- Energy can be exchanged with the electrons in an atom but only in **quanta** (packets of energy) that match the difference in energy levels of the atom



Photon energy $E = h\nu$

Energy difference between electron states

$$\Delta E = E_x - E_y$$

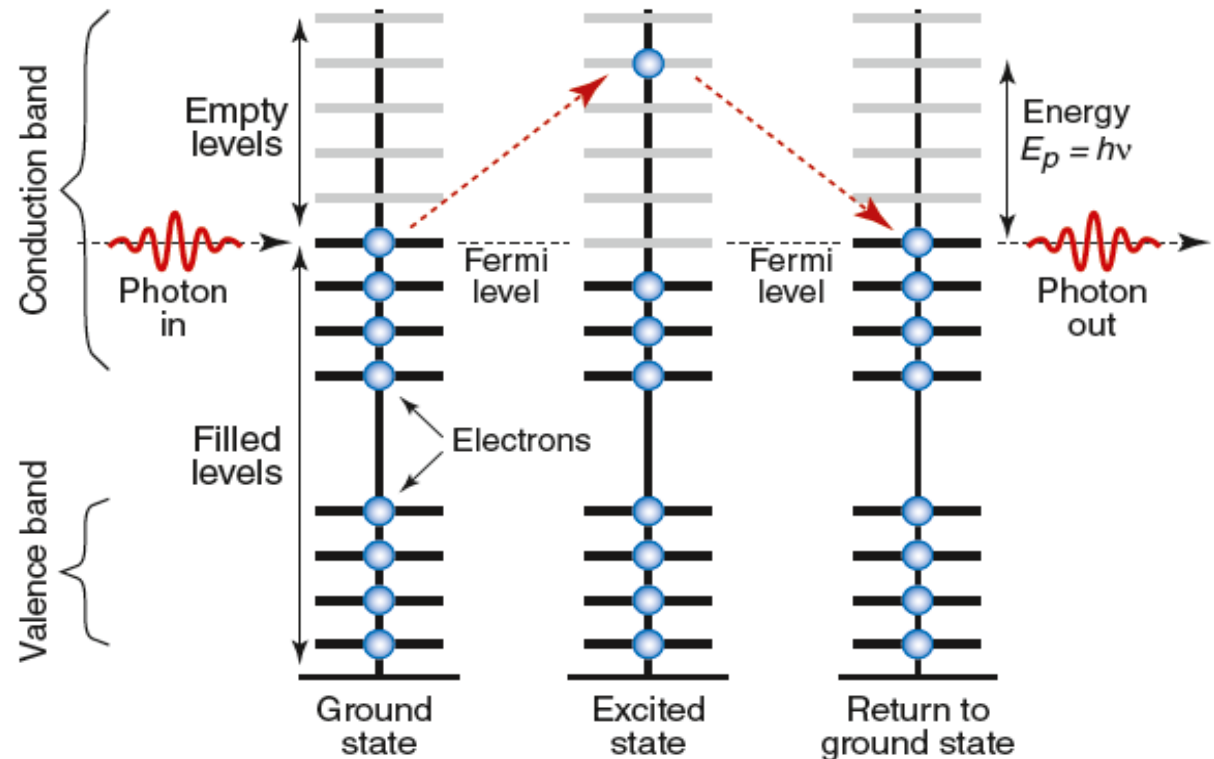
Electron excited from its ground state
when $h\nu \geq \Delta E$

Excited electron will return to its
ground state emitting $\Delta E = h\nu$
but could be by different transitions

Optical properties of metals

• Reflection in metals

- Metals absorb photons, capturing their energy by promoting an electron from the filled part of the conduction band into a higher, empty level
- When the electron falls back, a photon is re-emitted
- Emission of light energy results in reflection



Optical properties of metals

- **Reflection in metals**

- **Reflectivity** = I_R/I_0 is between 0.90 and 0.95 (1 for silver, Ag).
- Metal surfaces appear shiny
- Most of absorbed light is reflected at the same wavelength
- Small fraction of light may be absorbed
- Color of reflected light depends on wavelength distribution
 - Example: The metals copper and gold absorb blue and green light => reflected light has gold color



Optical properties of non-metals

- **Reflectivity of non-metals**
 - For normal incidence and light passing into a solid having an index of refraction n :

$$R = \text{reflectivity} = \left(\frac{n - 1}{n + 1} \right)^2$$

- Example: For Diamond $n = 2.41$

$$R = \left(\frac{2.41 - 1}{2.41 + 1} \right)^2 = 0.17$$

\ 17% of light is reflected

Optical absorption

- **Light Absorption – Beer's Law**

The amount of light absorbed by a material can be calculated using **Beer's Law**

$$I'_T = I'_0 e^{-\beta \ell}$$

Rearranging and taking the natural log of both sides of the equation leads to

β = absorption coefficient of material, cm^{-1}

ℓ = sample thickness, cm

I'_0 = incident light intensity

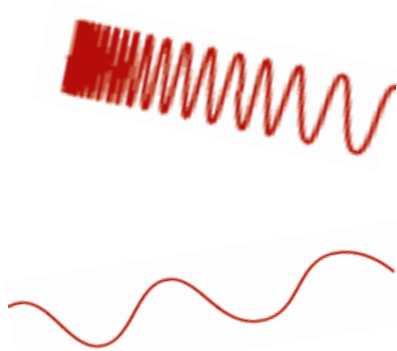
I'_T = transmitted light intensity

$$\ln \left[\frac{I'_T}{I'_0} \right] = -\beta \ell$$

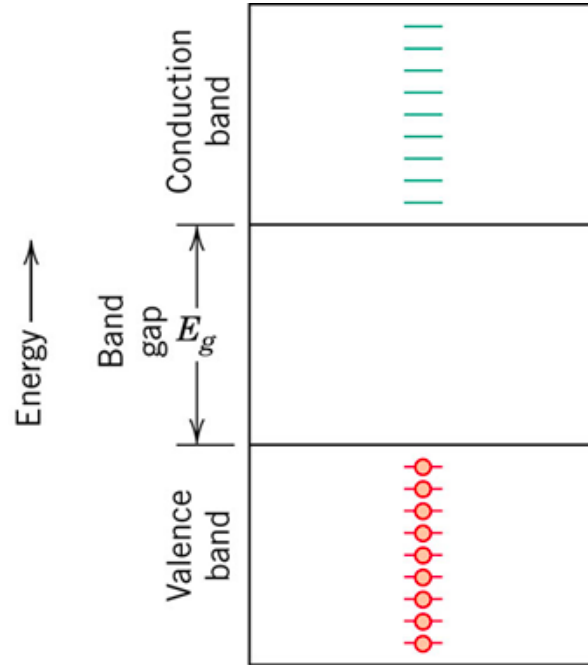
Beer's Law is true for all electromagnetic radiation, not just light
e.g. Infra-red, ultra-violet, X-rays

Optical absorption in insulators

Short wavelength light
High energy



Long wavelength light
Low energy



$$E_{\text{photon}} > E_{\text{gap}}$$

Absorption occurs

However, in most cases:

$$E_{\text{photon}} < E_{\text{gap}}$$

No absorption

**Insulators have such a large band gap that there are no optical frequencies of light that have enough energy for the process to occur
Many are therefore transparent**

Optical absorption in insulators

- **Color determined by the distribution of wavelengths:**
 - transmitted light
 - re-emitted light from electron transitions
- **Example 1:** Cadmium Sulfide (CdS), $E_g = 2.4 \text{ eV}$
 - absorbs higher energy visible light (blue, violet)
 - color results from red/orange/yellow light that is transmitted
- **Example 2:** Ruby = Sapphire (Al_2O_3) + (0.5 to 2) at% Cr_2O_3
 - Sapphire is transparent and colorless ($E_g > 3.1 \text{ eV}$)
 - adding Cr_2O_3 :
 - alters the band gap
 - blue and orange/yellow/green light is absorbed
 - red light is transmitted
 - Result: Ruby is deep red in color

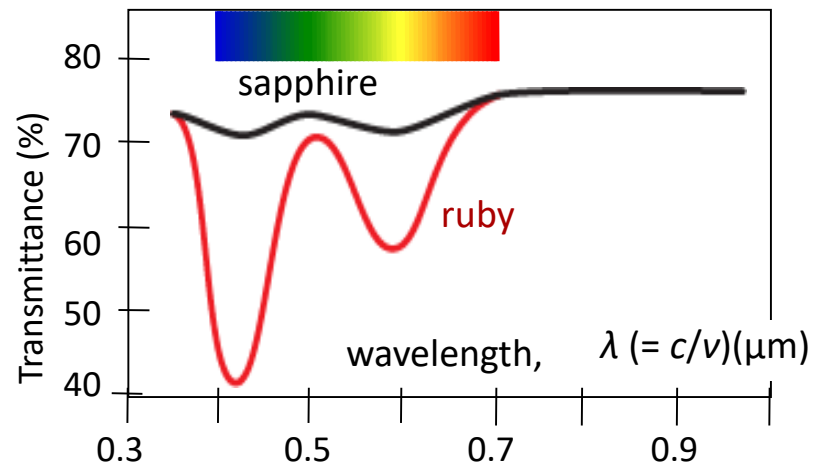
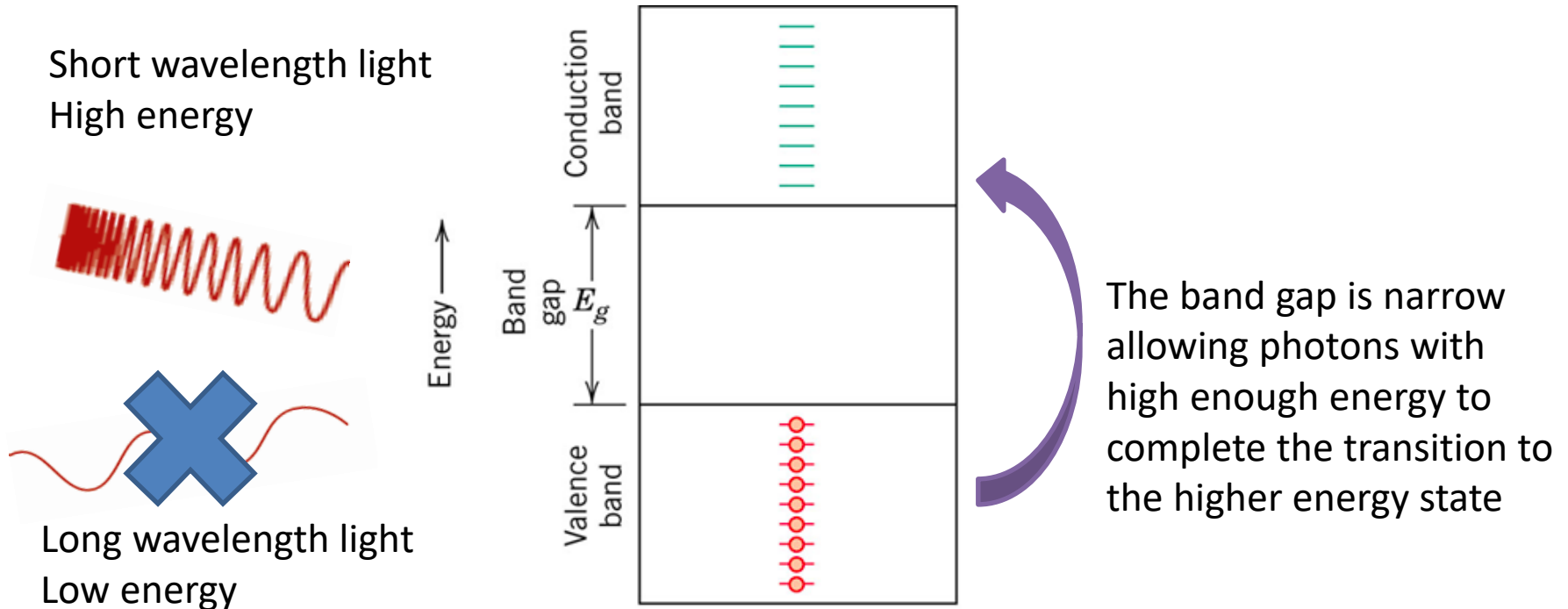


Fig. 22.9, Callister & Rethwisch 9e.

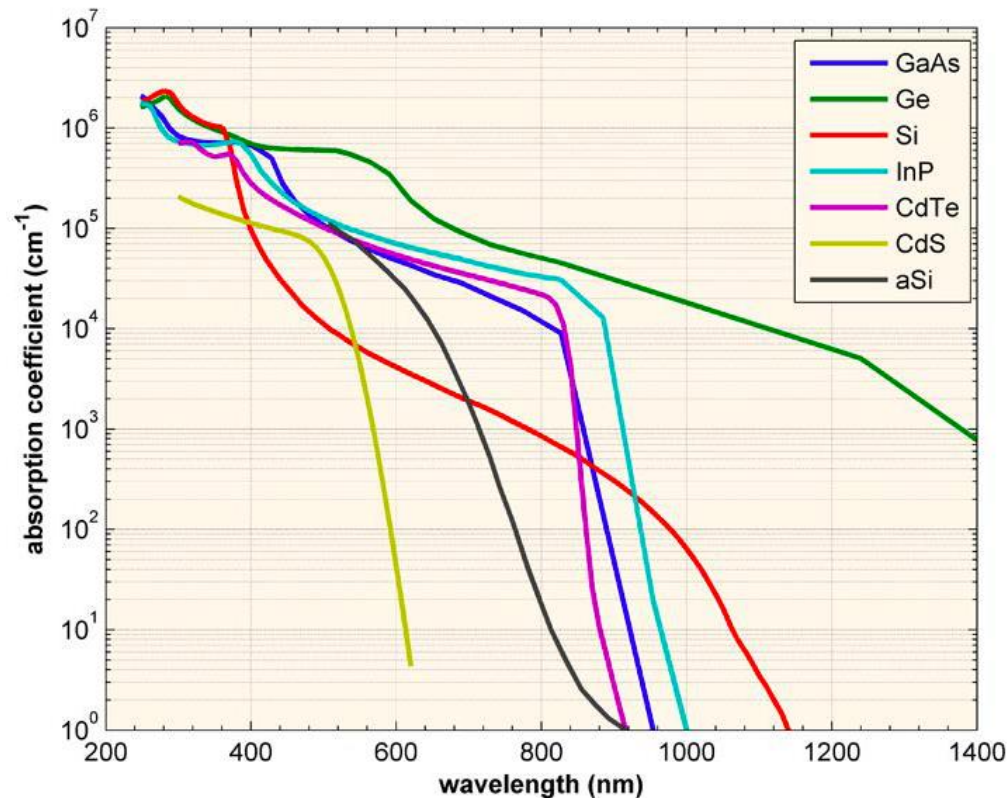
Optical absorption in semi-conductors



- If $E_{\text{gap}} < 1.8 \text{ eV}$, all light absorbed; material is opaque (e.g., Si, GaAs)
- If $E_{\text{gap}} > 3.1 \text{ eV}$, no light absorption; material is transparent and colorless (e.g., diamond)
- If $1.8 \text{ eV} < E_{\text{gap}} < 3.1 \text{ eV}$, partial light absorption; material is colored

Optical absorption in semi-conductors

- Semiconductors can absorb light at some wavelengths



2eV

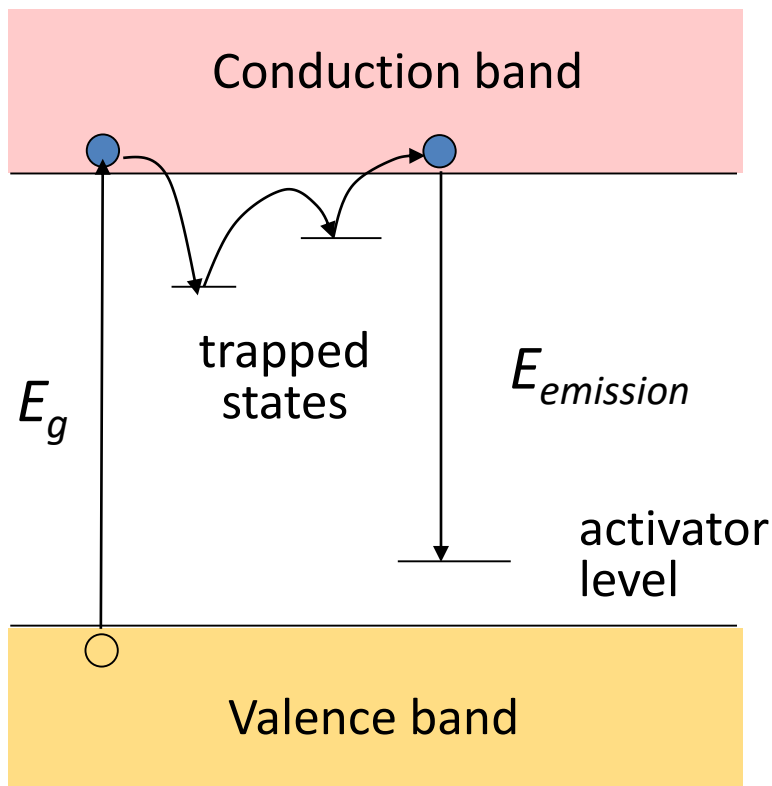
0.7eV



Visible spectrum

Optical luminescence

- **Luminescence** – reemission of light by a material
 - Material absorbs light at one frequency and reemits it at another (lower) frequency.
 - Trapped (donor/acceptor) states introduced by impurities/defects



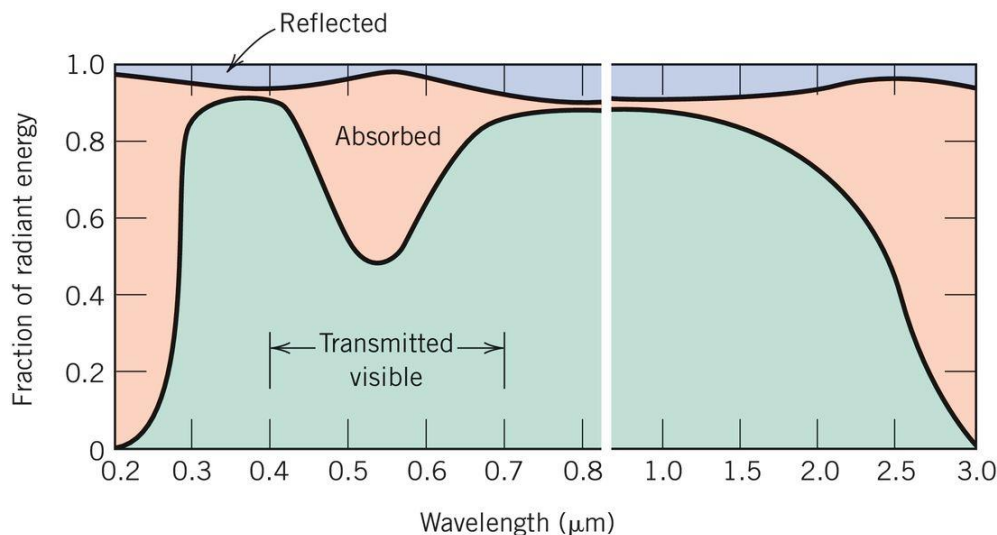
- If residence time in trapped state is relatively long ($> 10^{-8}$ s)
-- **phosphorescence**
- For short residence times ($< 10^{-8}$ s)
-- **fluorescence**

Example: Toys that glow in the dark. Charge toys by exposing them to light. Reemission of light over time—phosphorescence

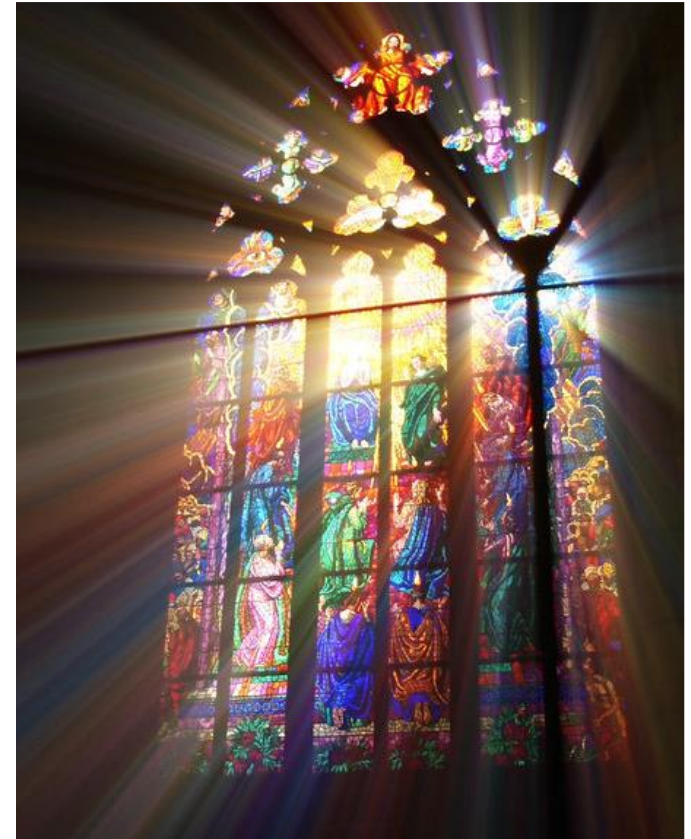
Optical transmission

• Transmission of light

- Passage of light through a transparent solid
- The fraction of incident light that is transmitted depends on the amount of light absorbed and reflected in the process
- The intensity of the transmitted light also depend on the thickness of the material

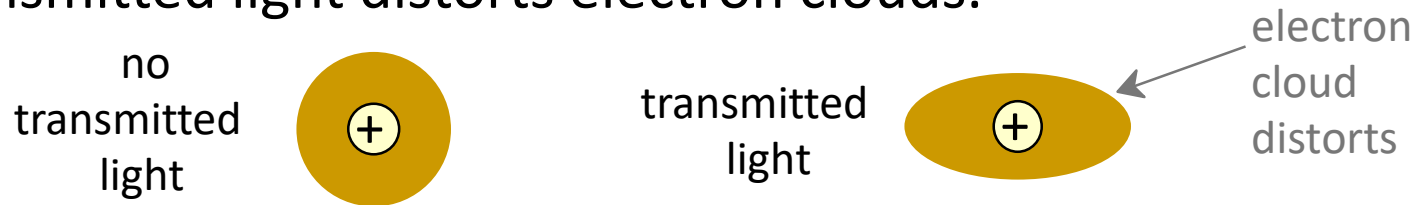


Example: green coloured glass



Refraction

- Transmitted light distorts electron clouds.



- The velocity of light in a material is lower than in a vacuum.

$$n = \text{index of refraction} \equiv \frac{c \text{ (velocity of light in vacuum)}}{v \text{ (velocity of light in medium)}}$$

$$n = \sqrt{\epsilon_r \mu_r} \cong \sqrt{\epsilon_r}$$

where

ϵ_r = relative permittivity (dielectric constant)

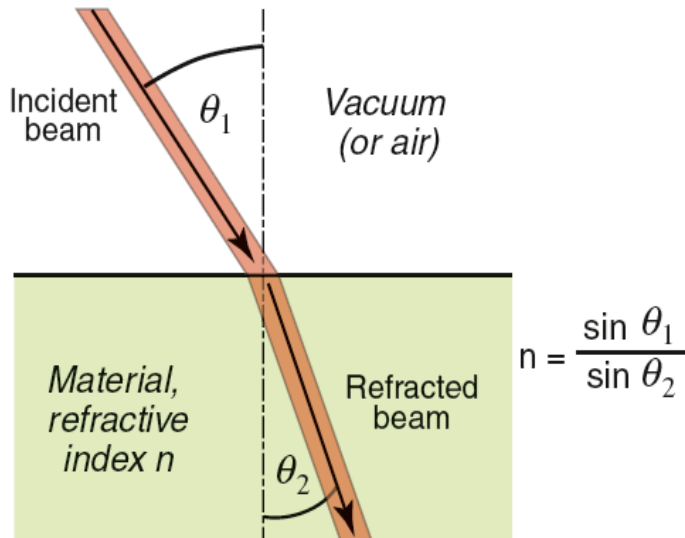
μ_r = relative permeability

Because the permeability of non-ferromagnetic materials is small
The refractive index of a material is dominated by the permittivity

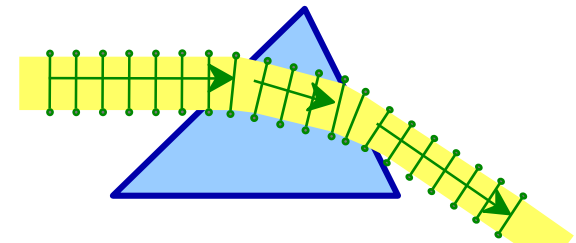
Refraction

Material	n
Typical glasses	1.5 -1.7
Plastics	1.3 -1.6
PbO (Litharge)	2.67
Diamond	2.41

Selected values from Table 22.1, *Callister & Rethwisch 9e*.



- Adding large ions (e.g., **Pb**) to glass decreases the speed of light in the glass.
- Light can be “bent” as it passes through a transparent prism



Summary



Optical properties:

Interaction of light waves with the electronic structure of solids

Light can be:

- Reflected
- Absorbed
- Luminescent
- Transmitted
- Refracted
- Scattered (in lecture 2)